ScootMeal (Food Ordering Platform)
Requirements Specification

1. Scope and users

1.1 The three applications

Application	Used by	Main purpose
Customer App	Customers (iOS + Android)	Browse and order food, dine-in, book tables, pre-order, track delivery, review, use vouchers
Merchant App	Restaurants / stalls (iOS + Android)	Receive order and reservation pop-ups, accept / reject, manage menu, hours, promotions, analytics
Web Admin	Platform administrators (browser)	Approve merchants, manage users, monitor orders and reservations, resolve disputes, moderate content

1.2 In scope

- Customer ordering: delivery, dine-in, takeaway, and mixed dine-in + delivery in one order
- Dine-in table reservations with optional pre-order, approved by the restaurant
- Merchant-side order and reservation handling through timed pop-ups
- Web admin panel for platform management
- Stripe payments; delivery through a third-party provider API (simulated now, Lalamove planned)
- Publishing the Customer and Merchant apps to Google Play and the Apple App Store

1.3 Later (Could Have) and out of scope

- Referral programme, late-night order discounts, agentic AI ordering, merchant review responses (Could Have)
- Out of scope for this phase:
  - In-house rider / courier management (a vendor API is used instead)
  - Real-time traffic-aware routing
  - Calorie tracking and long-term dietary memory
  - Auto-payment through AI

2. Functional Requirements

Priority: M = Must, S = Should, C = Could. Source: UR- / MR- = earlier requirement list; Client = new client request; Prototype = already built; Research = Research Report; Proposed = team proposal to confirm.

2.1 Customer App - account and discovery

ID	Requirement	Pri.	Source
FR-1.1	The system shall let a customer create an account and log in; an account is required to place an order (no guest checkout).	M	UR-1
FR-1.2	The system shall collect date of birth at sign-up and block registration below the platform's minimum age.	M	MoSCoW
FR-1.3	The system shall let a customer browse restaurants and menus, and search food by dish type (e.g. "beef noodle") across all merchants.	M	UR-2, UR-3
FR-1.4	The system shall let a customer filter dishes by dietary need (spicy, allergy-free, vegetarian, plant-based) and style (dry / wet).	S	UR-4
FR-1.5	The system shall show each merchant's distance from the customer's chosen delivery address, and let the customer set an address different from their current location.	M	UR-5, UR-6
FR-1.6	The system shall show whether each restaurant is currently open, based on its operating hours.	M	UR-7
FR-1.7	The system shall provide a trending section (food, drinks, desserts, cafes) and a section for underground / hidden-gem cafes.	C	UR-18, UR-19, UR-20

2.2 Customer App - ordering and fulfilment

ID	Requirement	Pri.	Source
FR-2.1	The system shall let a customer add items to a cart and pay securely through Stripe.	M	UR-8
FR-2.2	The system shall let a customer choose Delivery, Dine-in or Takeaway for an order.	M	UR-9
FR-2.3	The system shall allow one order to mix fulfilment modes: each item (or quantity) in the cart is set to Delivery or Dine-in (e.g. 2 dishes dine-in and 3 dishes delivery), and the whole order is paid in one Stripe payment.	M	Client
FR-2.4	The system shall let a customer request simple customisations (e.g. less spicy) where the merchant supports them.	S	UR-11
FR-2.5	The system shall notify the customer in real time if a merchant marks an ordered item out of stock, and let the customer accept a partial order or cancel for a full refund.	M	UR-12
FR-2.6	The system shall let the customer track order status and, for delivery items, the rider's live location.	M	UR-13
FR-2.7	When the merchant marks dine-in items as Ready, the Customer App shall show a pop-up ("Your food is ready - please collect") and make the phone vibrate, with a push notification if the app is in the background. The pop-up stays until the customer confirms collection.	M	Client
FR-2.8	The system shall let a customer rate and review individual food items.	M	UR-14
FR-2.9	The system shall let a customer use vouchers, loyalty points and promotions at checkout.	S	UR-15

2.3 Customer App - reservations and pre-order

ID	Requirement	Pri.	Source
FR-3.1	The system shall let a customer book a table by choosing a venue (browsable by category: Trending, Buffet, Luxury, Aesthetic Cafe), date, time slot and party size.	S	UR-10, Prototype
FR-3.2	The system shall let a customer add a pre-order of dishes to a reservation before submitting it.	M	Client
FR-3.3	The system shall let a customer choose Pay Now or Pay Later for a reservation.	S	Prototype
FR-3.4	After submission the system shall show the reservation as "Awaiting restaurant" and notify the customer when it is accepted, declined or expired, and separately when the pre-order is approved or declined.	M	Client
FR-3.5	The system shall let a customer cancel or modify a reservation up to a configurable cut-off before the slot, and send a reminder before each confirmed reservation.	S	Proposed

2.4 Customer App - engagement

ID	Requirement	Pri.	Source
FR-4.1	The system shall let a customer browse F&B job listings posted by merchants.	S	UR-16

FR-4.2	The system shall provide an AI assistant, by text or voice in English, Malay or Mandarin, to help find food and track orders.	S	UR-17

2.5 Merchant App

ID	Requirement	Pri.	Source
FR-5.1	The system shall let a merchant register and set up a business profile through a short onboarding process with minimal paperwork; the account becomes active once a web admin approves it.	M	MR-1, MR-13
FR-5.2	The system shall let a merchant add, edit and remove menu items and prices, and list items under food-type categories.	M	MR-2, MR-3
FR-5.3	The system shall let a merchant suspend a category or item temporarily (e.g. sold out) without deleting it, and reactivate it later.	M	MR-4
FR-5.4	The system shall let a merchant set which customisations each dish accepts, and set operating hours that drive the open / closed status.	S	MR-5, MR-6
FR-5.5	When a paid order arrives, the Merchant App shall show a full-screen pop-up with an alert sound, the items and their fulfilment modes, Accept and Reject buttons, and a 10-minute countdown.	M	MR-7, Client
FR-5.6	The system shall let a merchant mark items out of stock in real time, which notifies affected customers immediately.	M	MR-8
FR-5.7	The system shall let a merchant mark dine-in items Ready to collect (which triggers FR-2.7) and hand over delivery items to the rider.	M	Client
FR-5.8	When a customer submits a reservation, the Merchant App shall show a pop-up with the booking details and any pre-order, with Approve / Decline buttons and a 30-minute countdown.	M	MR-9, Client
FR-5.9	The reservation pop-up shall include a Call customer button that opens the phone dialer with the customer's number, so the restaurant can liaise before replying.	M	Client
FR-5.10	The restaurant shall be able to approve or decline the pre-order separately from the reservation (e.g. accept the table but decline a dish it cannot prepare).	M	Client
FR-5.11	The system shall let a merchant create promotions, vouchers and "purchase with purchase" deals.	S	MR-10
FR-5.12	The system shall show a merchant sales analytics (revenue, best-selling items) for a chosen period.	S	MR-11
FR-5.13	The system shall let a merchant post job vacancies, and (later) respond to reviews.	S	MR-12

2.6 Web Admin

ID	Requirement	Pri.	Source
FR-6.1	The system shall provide a web dashboard with admin-only login and role-based access.	S	Proposed
FR-6.2	The system shall let an admin review, approve, reject or suspend merchant registrations.	S	Proposed
FR-6.3	The system shall let an admin view and suspend customer accounts, including accounts that fail the age check.	S	Proposed
FR-6.4	The system shall let an admin view all orders and reservations and, for disputes, cancel them or trigger a Stripe refund.	S	Proposed
FR-6.5	The system shall let an admin moderate reviews and job listings, and manage platform-wide promotions and trending sections.	S	Proposed
FR-6.6	The system shall show platform analytics (orders, revenue, merchant performance).	S	Proposed

2.7 Logistics and Payments

ID	Requirement	Pri.	Source
FR-7.1	The system shall integrate with a third-party delivery API to get quotations, book riders and receive status updates by webhook. A simulated provider is used in this phase; Lalamove is the planned provider.	M	MoSCoW, Research
FR-7.2	Only the delivery items of an order shall be sent to the delivery provider; dine-in items stay with the merchant.	M	Client
FR-7.3	The system shall request a new rider if none accepts, and shall be able to fall back to Grab Express if Lalamove cannot serve a request.	S	Research

FR-7.4	The system shall pass delivery status changes and the rider's live location to the Customer and Merchant apps within 5 seconds of receiving them.	M	UR-13
FR-7.5	The system shall record every payment with a unique reference and shall automatically refund in full when an order is rejected or times out, a reservation is declined or expires, or a customer cancels a partial order.	M	Client

3. Pop-ups and time limits

These are the timed interactions the client asked for. They are shown in Figures 2–4.

Event	Who responds	Time limit	If there is no response / outcome
New paid order	Merchant	10 minutes	Order is auto-cancelled, full refund, customer notified
New reservation request	Merchant	30 minutes	Request expires, any payment refunded, customer notified
Pre-order inside a reservation	Merchant	Same pop-up as the reservation	Merchant approves or declines it separately; a declined pre-order does not cancel the reservation
Dine-in food ready	Customer	No time limit	Pop-up and buzz stay until the customer confirms collection

4. Non-Functional Requirements

ID	Category	Requirement
NFR-1	Performance	95% of API requests (search, checkout, order placement) shall complete within 2 seconds under normal load.
NFR-2	Timers	Pop-ups shall reach the Merchant App within 5 seconds of the triggering event. Countdowns shall run on server time so they stay correct if the app is closed and reopened.
NFR-3	Reliability	Failure of the delivery provider shall trigger retry or reassignment, never a silently failed order.
NFR-4	Availability	The system shall target 99% uptime during operating hours (8am-10pm), excluding scheduled maintenance.
NFR-5	Security	All traffic shall use HTTPS/TLS. Card data is handled only by Stripe (tokenised) and is never stored by ScootMeal. Customer, merchant and admin roles shall be separated by role-based access control.
NFR-6	Usability	Core customer flows (search to paid order, or venue to submitted reservation) shall take no more than 5 screens from Home.
NFR-7	Scalability	The API layer shall be stateless so more instances can be added as order volume grows.
NFR-8	Maintainability	The backend shall be modular (Auth, Orders, Reservations, Merchant Management, Logistics, Promotions, Admin) so a module such as the delivery provider can be replaced independently.
NFR-9	Accessibility	Dietary and Halal badges shall be distinguishable by icon or text, not colour alone.
NFR-10	Deployment	The Customer and Merchant apps shall be publishable to Google Play and the Apple App Store.

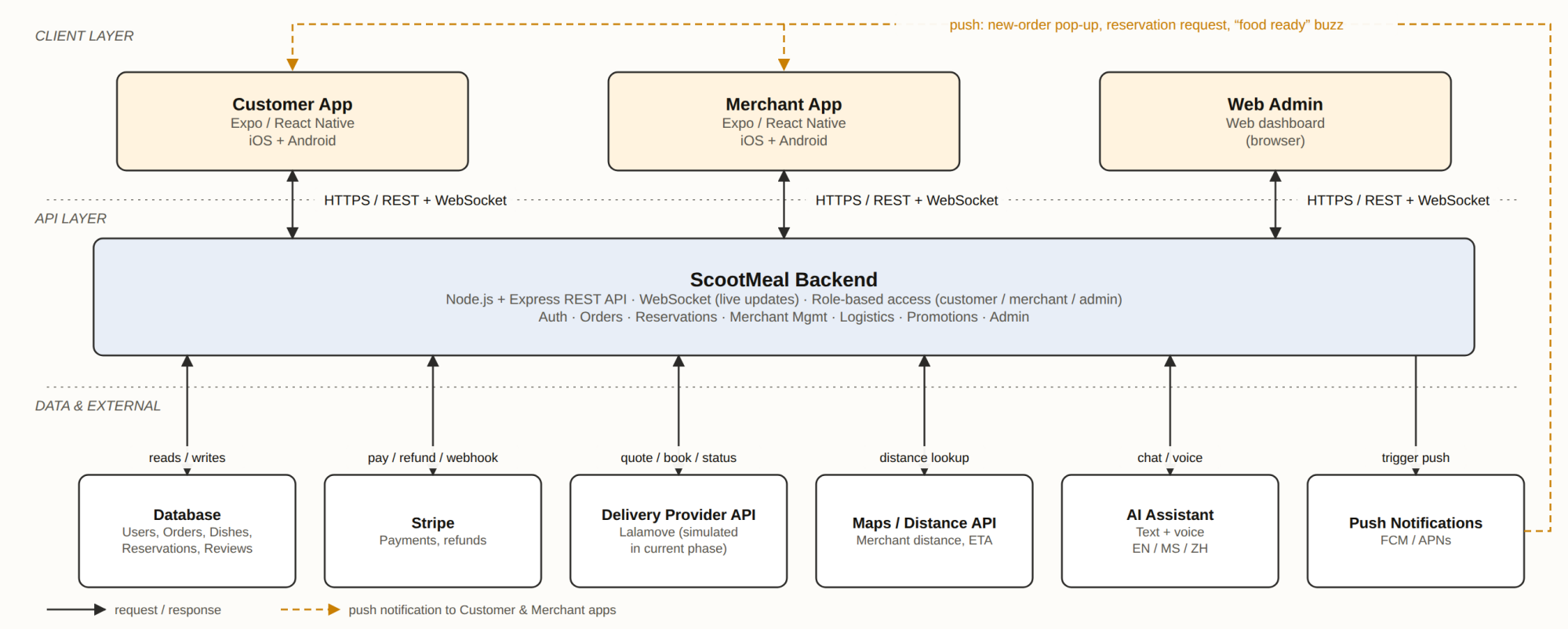
5. Assumptions to confirm

Points where the brief was not specific. They are written into the diagrams, so please confirm or correct them.

1. The "10-minute pop-up" is taken to mean the merchant's accept / reject window for a new order, with auto-cancel and refund on timeout. Please confirm this is what the client meant, and what should happen on timeout.
2. The reservation pop-up expires after 30 minutes with no reply; the customer is told and any payment is refunded. A declined pre-order does not cancel the reservation.

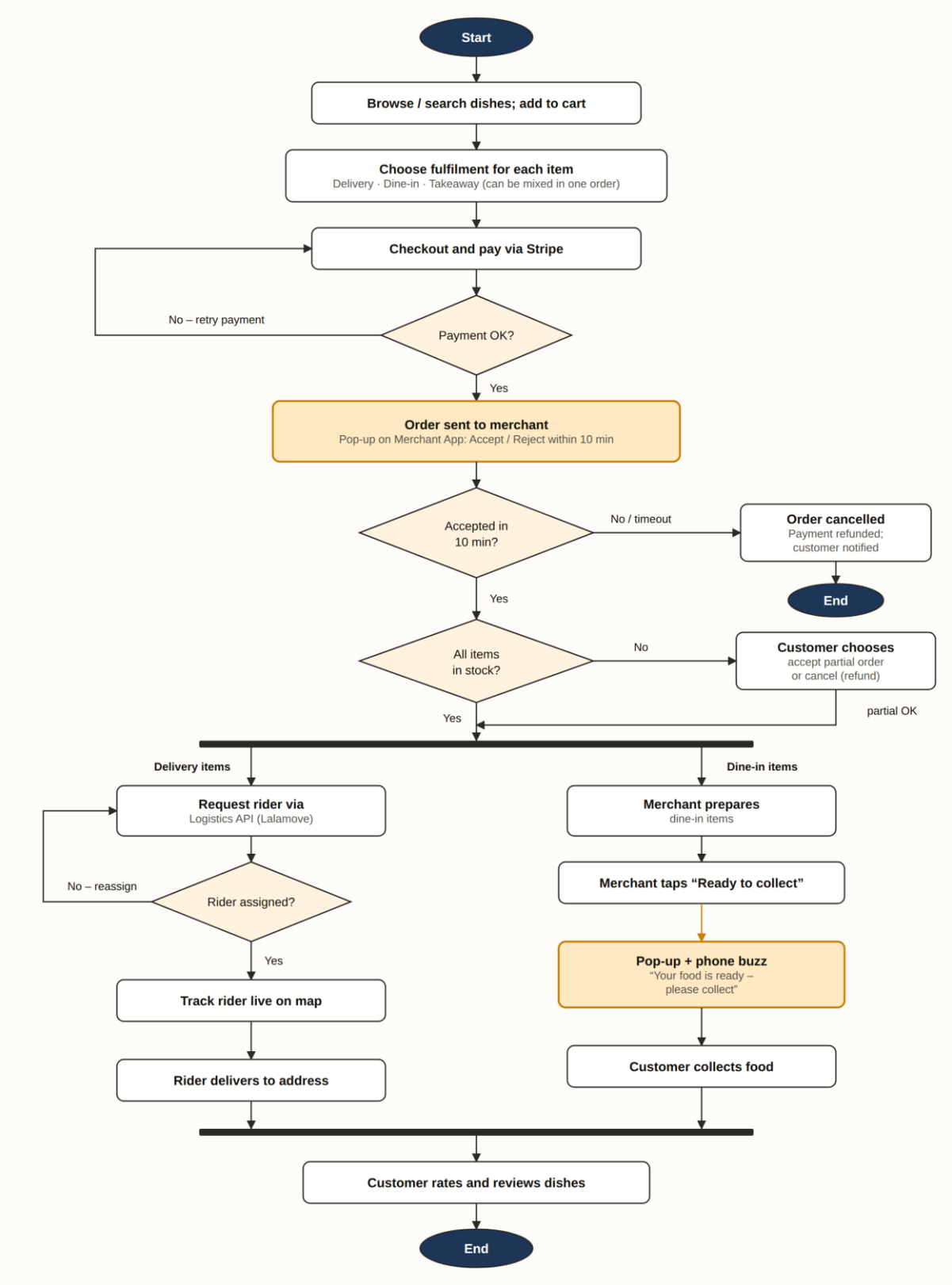
3. A Pay Now reservation payment is refunded in full if the restaurant declines or does not reply.
4. Customers must log in before checkout (earlier requirement: no guest checkout). Browsing before login is left open.
5. Delivery is simulated in this phase, as in the earlier MoSCoW list; the Lalamove integration follows the Research Report.
6. The Web Admin requirements (Section 3.6) are proposed by the team and were not in the earlier requirements list or the client brief.

Figure 1. System Architecture



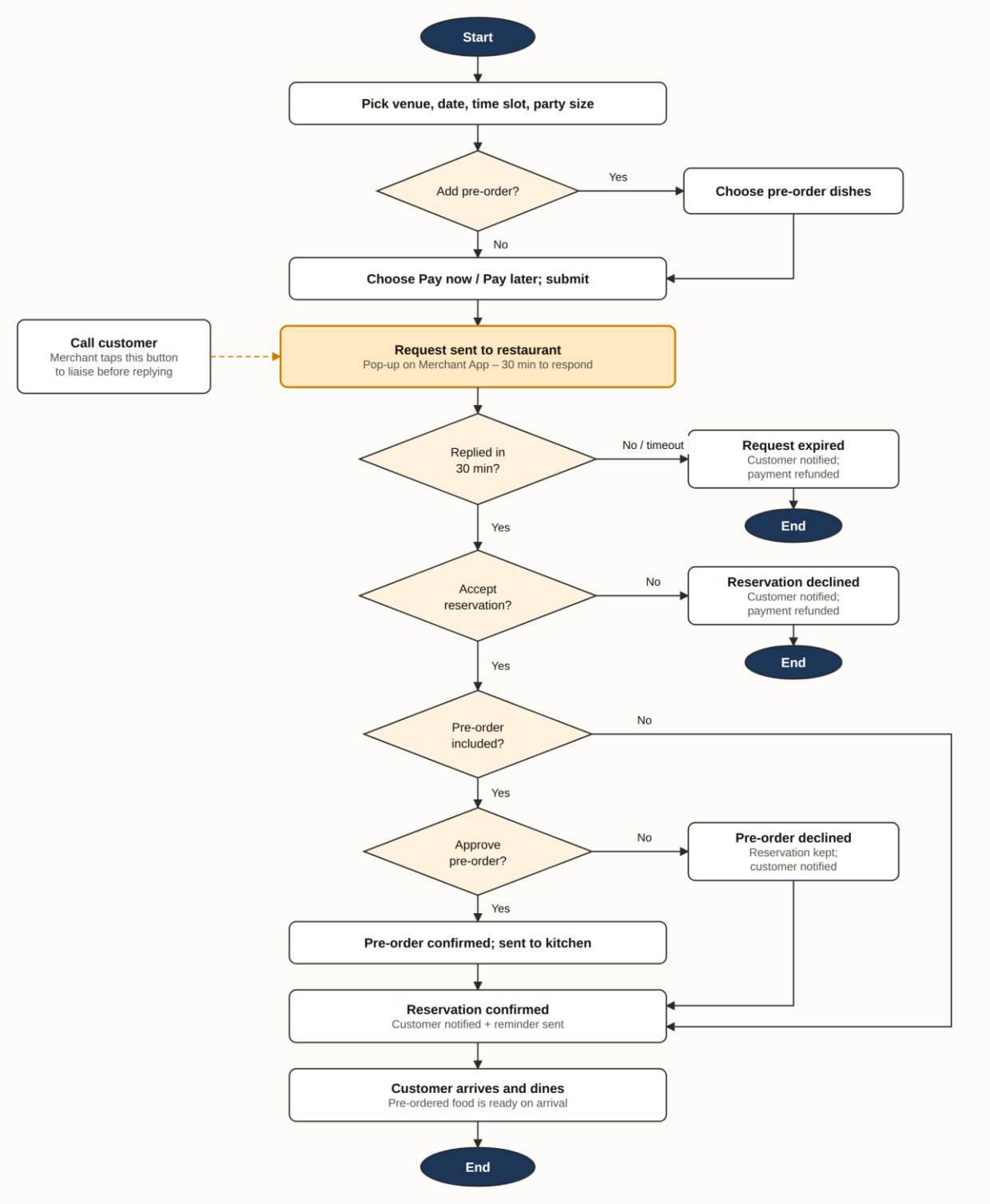
Three clients (Customer App, Merchant App, Web Admin) talk to one backend over HTTPS/REST and WebSocket. The backend uses the database and calls Stripe, the delivery provider, maps, the AI assistant and the push service; push notifications (dashed) deliver the order, reservation and “food ready” pop-ups to the apps.

Figure 2. Order Flowchart



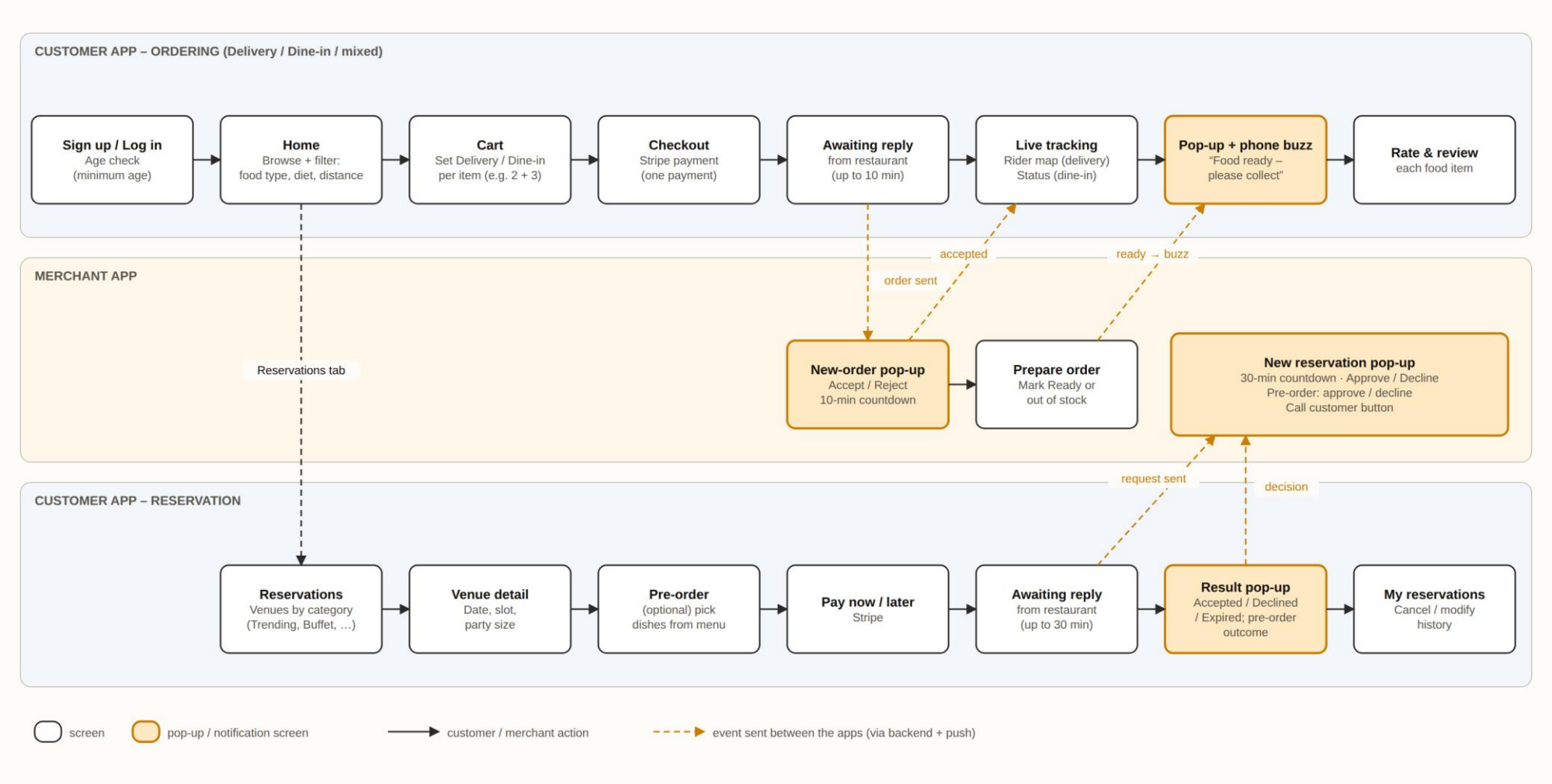
From cart to review. The merchant has 10 minutes to accept (Merchant App pop-up). After acceptance the order forks: delivery items go to the delivery provider, dine-in items are prepared and trigger the "food ready" pop-up and phone buzz; both join before the review.

Figure 3. Reservation and Pre-order Flowchart



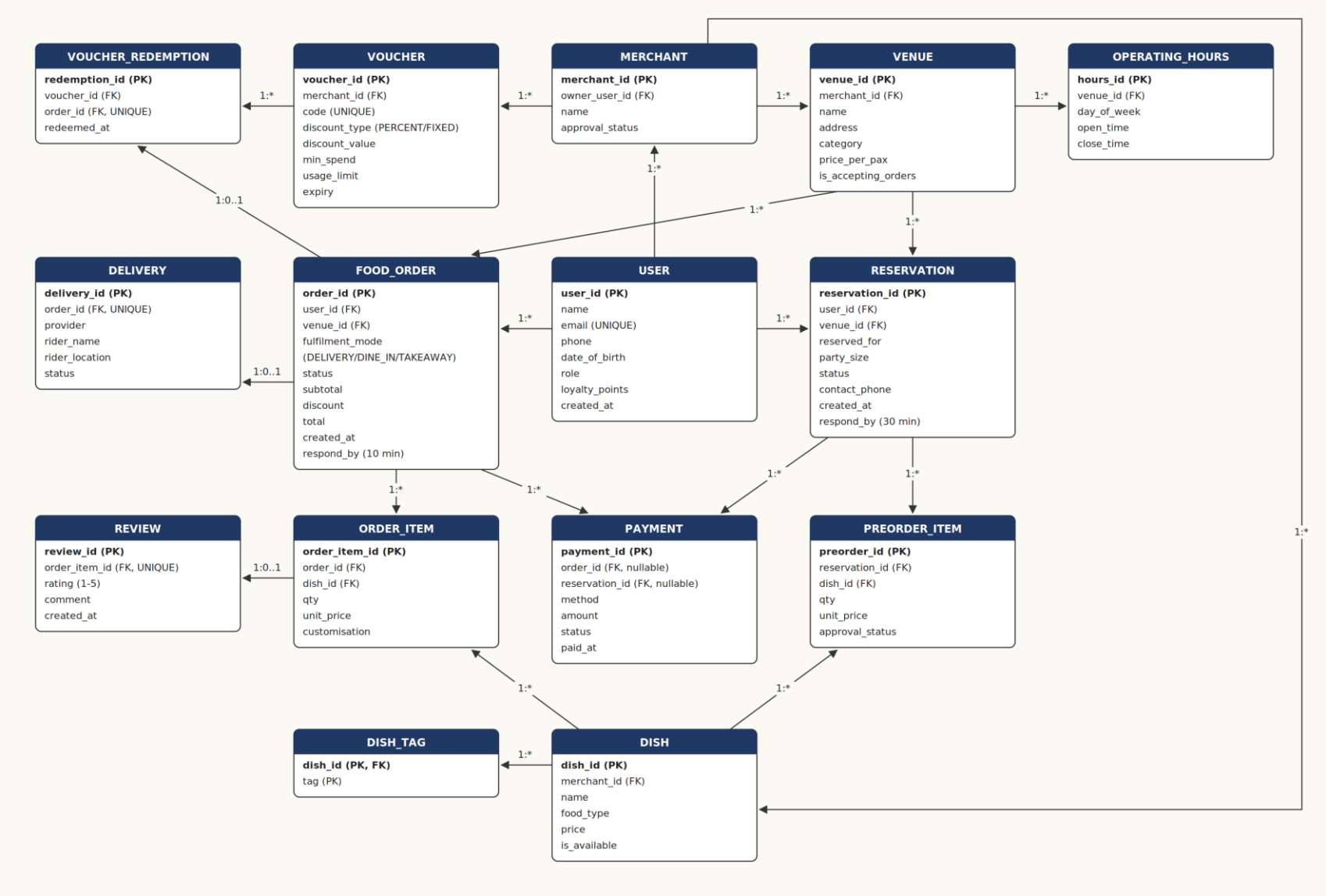
The restaurant has 30 minutes to reply to the reservation pop-up and can tap Call customer first. A pre-order is approved or declined separately; an unanswered or declined request is refunded.

Figure 4. User Flow



Customer ordering (top), Merchant App pop-ups (middle) and customer reservation (bottom). Orange boxes are pop-up screens; dashed orange arrows are events passed between the apps.

Figure 5. Entity-Relationship Diagram



Eleven entities. Each order item carries its own fulfilment mode (DELIVERY / DINE_IN / TAKEAWAY), so one order can mix dine-in and delivery. Orders and reservations store the response deadline (10 / 30 minutes).